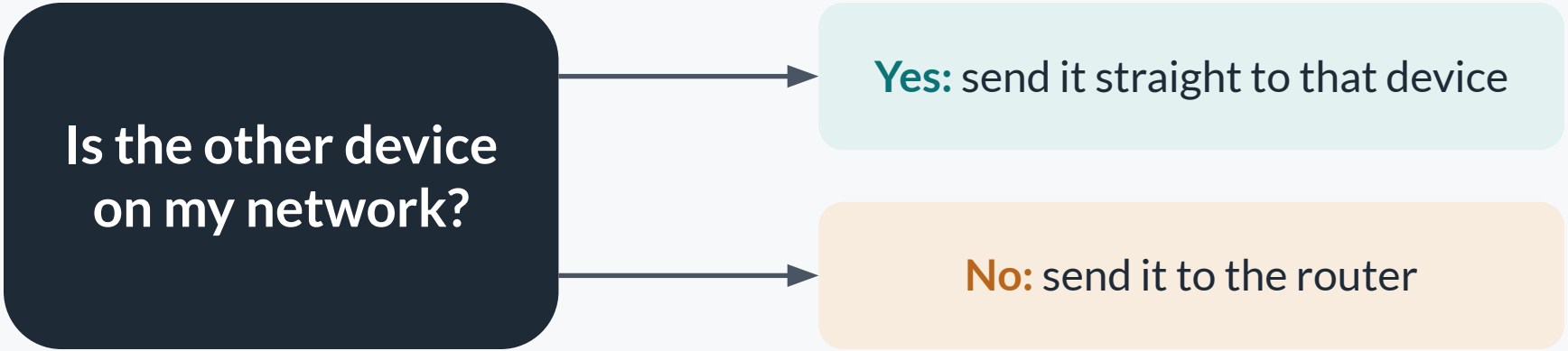
How Your Computer Finds Its Network

Using bitwise operators on IP addresses

Saaketh Srikakolapu, COMP 211

Every message starts with one question



Your computer answers this with **a bitwise AND**.

An IP address is just a 32-bit number

192	**168**	**1**	**37**
`11000000`	`10101000`	`00000001`	`00100101`

Each part is 8 bits, so the whole address is 4 × 8 = 32 bits.
In C, shifts and OR put the 4 parts side by side:

```
ip = make_ip(192, 168, 1, 37);
```

A subnet mask splits an IP into two parts

Mask	255	255	255	0
Binary	11111111	11111111	11111111	00000000
Part	network	network	network	host

1s mark the network part: which network you are on.
0s mark the host part: which device on that network.
Short way to write this mask: /24 (it starts with 24 ones).

AND with the mask gives the network address

IP	11000000	10101000	00000001	00100101	192.168.1.37
Mask	11111111	11111111	11111111	00000000	255.255.255.0
IP & Mask	**11000000**	**10101000**	**00000001**	**00000000**	**192.168.1.0**

Where the mask is 1, the IP's bit stays the same.
Where the mask is 0, the bit becomes 0.
In C: `network = ip & mask;`

Network and broadcast are mirror images

	In C	Host bits become	Result
Network address	`ip & mask`	all 0s	**`192.168.1.0`**
Broadcast address	`ip | ~mask`	all 1s	**`192.168.1.255`**

~mask flips every bit, so its 1s line up with the **host part**.
Every device on the network has an address between these two.
Usable addresses on a /24: 2^8 − 2 = 254

Are 192.168.0.200 and 192.168.1.5 on the same network?

The mask is 255.255.255.0 (/24)

One bit in the mask changes the answer

	/24 mask (255.255.255.0)	/23 mask (255.255.254.0)
Network of 192.168.0.200	`192.168.0.0`	`192.168.0.0`
Network of 192.168.1.5	`192.168.1.0`	`192.168.0.0`
Same network?	No	Yes

Third part: 0 = 00000000, 1 = 00000001. Only the last bit differs.
The /23 mask has a 0 in that spot, so AND wipes it out.
Smaller slash number = bigger network (/24: 254 usable, /23: 510).

Takeaways

- An IP address is a 32-bit number, so bitwise operators fit it well
- AND with the mask finds the network; OR with ~mask finds the broadcast
- Checking "same network?" is two ANDs and a compare
- Moving one bit between network and host doubles or halves the network

Questions?

Code and README: github.com/saakethsri/bitwise-subnets